

Finite-Memory Projection Corrections In Cosmological Consistency Diagnostics

Abstract

We introduce a bounded finite-memory projection operator for diagnostic cosmological consistency tests. The operator is formulated as an effective projection response over normalized depth, $w(x)=1+\rho x^3$, with a passive memory-budget bound $\rho \leq 4$. The construction is not presented as a complete cosmological model or an observational detection, but as a disciplined diagnostic module for comparing projected response shapes against backreaction-aware distance consistency reconstructions. In a BAO-only diagnostic reconstruction, the locked K2 window overlaps the full target envelope. In an SN+BAO source-split stress test, strict median-sign matching exhibits a localized tension; a reconstruction-aware sign-stability gate resolves this tension without widening the K2 window. The result motivates a future full-likelihood comparison while preserving clear claim boundaries.

1. Introduction

Cosmological consistency tests increasingly distinguish background expansion, light-propagation effects, and cosmic backreaction. A residual relative to a standard homogeneous model is therefore not automatically evidence for new physics: it may arise from optical sampling, reconstruction convention, or averaging effects. This note develops a deliberately narrow projection-memory diagnostic that can be placed alongside such tests.

The objective is not to disclose or validate a full underlying theory. Instead, we ask whether a predeclared, bounded projection-memory operator can define a reproducible diagnostic window and survive basic source-split stress tests.

2. Finite-Memory Operator

Let $x \in [0, 1]$ denote normalized projected depth. We consider a finite-memory response multiplier

$$w(x) = 1 + \rho x^3.$$

The passive memory-budget condition requires the accumulated tail response not to exceed the unit local response budget:

$$\int_0^1 \rho x^3 dx \leq \int_0^1 1 dx.$$

Thus

$$\begin{aligned} \rho/4 &\leq 1, \\ \rho &\leq 4. \end{aligned}$$

The cubic power is selected from the power-kernel family $w_p(x)=1+\rho x^p$. Lower powers are too visible at low depth, while higher powers concentrate the memory budget too sharply near the endpoint. The resulting locked diagnostic window is

$$\begin{aligned} \eta &\in [0.25, 0.50], \\ \rho &\in [3.00, 4.00]. \end{aligned}$$

3. Diagnostic Protocol

The diagnostic targets are reconstructed envelopes rather than direct published likelihoods. We therefore separate three levels of comparison:

sign-stable reconstructed points	-> exact sign gate
sign-unstable reconstructed points	-> envelope / diagonal residual gate
controls and extrapolations	-> stress diagnostics only

This policy prevents an unstable reconstructed median sign from acting as a hard falsifier while preserving the stricter sign requirement where the reconstruction is stable.

4. Preliminary Results

The locked K2 window overlaps the BAO-only diagnostic envelope at all tested points. In SN+BAO, strict median-sign matching fails at a localized point near $z=1.1925$. That point is method-degree sensitive: lower-degree reconstructions are positive, while higher-degree reconstructions pull the median negative.

Under the sign-stability-aware diagnostic gate, all stable-sign points pass the exact sign check and all unstable-sign points pass the envelope / diagonal residual check.

5. Limitations

This manuscript is a theory-method diagnostic note. It does not claim a full covariance likelihood, a direct observational detection, or a complete derivation of cosmology from an underlying action. The next stage is a paper-aligned likelihood comparison using covariance or shrinkage-covariance products and direct published likelihood data where available.